

Business Case

YAT LMS Replacement — GrayBoard to DOODLE

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1. Executive Summary

YAT's Learning Management System, GrayBoard, is a bespoke in-house platform that has reached end of supportable life. Its availability and performance have been degrading, it has no vendor support or product roadmap, and it does not meet current accessibility obligations — placing YAT's teaching delivery and compliance at growing risk. This Business Case evaluates two options over a five-year horizon: Option A — remediate and retain GrayBoard; Option B — replace it with a supported contemporary LMS (DOODLE) deployed on-premises at the Cremorne campus.

Option A is ~\$63,600 cheaper over five years on direct cost, but it only defers the problem: it leaves YAT on an unsupported, dead-end platform with a heavy in-house maintenance burden and a fragile accessibility retrofit, and a full replacement would still be required. Option B delivers a vendor-supported, WCAG 2.1 AA-compliant, performant LMS with an active product roadmap, preserves all student records, and actually reduces YAT's ongoing maintenance burden — partly offsetting the higher upfront cost.

We recommend Option B. We ask the board to approve the replacement, the phased implementation plan in §9 (with cutover scheduled into the December–January teaching break), and the Year-1 budget envelope.

2. Strategic Alignment Analysis

YAT's ICT direction in 2022 prioritises reliable, well-supported core systems; the capacity to support continued student growth; and meeting the College's statutory obligations as a Registered Training Organisation. The LMS is the platform through which teaching, assessment and student records are delivered and held, so its reliability and compliance are foundational to that direction.

Across the RTO sector, institutions have moved decisively off bespoke in-house LMS platforms onto established, vendor-supported products with active communities and accessibility conformance — driven by the maintenance cost and compliance risk of home-grown systems, and by rising accessibility expectations under the Disability Discrimination Act 1992. YAT's continued reliance on GrayBoard runs against this direction. Cloud-hosted delivery is the emerging next step in the sector; YAT notes it as a future consideration but, consistent with its then-current on-premises operating model, this engagement is scoped to an on-premises replacement. Replacing GrayBoard now with a supported product both resolves the immediate risk and positions YAT for a future move to cloud hosting.

3. Current State — the GrayBoard LMS

GrayBoard is a bespoke LMS built and maintained in-house by YAT ICT, running on on-premises infrastructure at the Cremorne campus and integrated with Active Directory (authentication), Office 365 (email notifications), and AVETMISS/ASQA statutory reporting. It has reached end of supportable life: effective availability has been degrading, typical and peak-load performance is declining (the platform struggles with the ~3× concurrent load during end-of-term assessment windows), and it depends on a small number of YAT staff for continued maintenance with no external support or product roadmap. It does not meet WCAG 2.1 Level AA, creating a compliance exposure under the Disability Discrimination Act. It does, however, hold YAT's complete student-record history, which carries long-tail retention obligations under the Standards for RTOs 2015 — so any replacement must preserve those records with full integrity. These limitations — no support, declining availability and performance, accessibility non-conformance, and concentration risk on in-house maintenance — are what drive the gap analysis.

4. Gap Analysis

Strategic objective	Current state (GrayBoard)	Gap	Improvement opportunity	Proposed change
Supported, current LMS platform	Bespoke, end-of-life, no vendor support/roadmap	No support or roadmap; concentration risk	Adopt an established commercial LMS	Replace with DOODLE (on-prem)
Availability $\geq 99\%$	Degrading, below target	Reliability shortfall	Modern supported stack	Deploy DOODLE on supported infrastructure
WCAG 2.1 AA accessibility	Non-conformant (DDA exposure)	Compliance gap	Product with native AA conformance	Replace with conformant LMS
Peak-load performance	Degrades at $\sim 3\times$ term-end load	Performance shortfall	Right-sized supported platform	Provision DOODLE to peak profile
Preserve all student records	Held in GrayBoard	Migration risk if not managed	Verified ETL migration	Migrate with reconciliation + parallel run

The gaps cluster around supportability, reliability, compliance and performance — none of which an in-house remediation truly closes, and all of which a supported replacement addresses while also preserving the student-record history.

5. Options Considered and Evaluation

5.1 Workload definition

The replacement must: be a current, vendor-supported LMS product; run on YAT's Windows Server 2016 on-premises hosting standard; preserve all GrayBoard student records with full integrity; achieve $\geq 99\%$ availability; meet WCAG 2.1 AA; match or exceed GrayBoard's performance (including $\sim 3\times$ term-end peaks); integrate with Active Directory, Office 365 and AVETMISS reporting; and carry vendor support with severity-1 response within 4 hours during teaching hours.

5.2 Options considered

- Option A — Remediate and retain GrayBoard: invest in an accessibility retrofit and a hardware refresh, and continue in-house maintenance of the legacy platform.
- Option B — Replace with a contemporary on-prem LMS (DOODLE): select an established product via market evaluation (see Appendix 2), deploy it on YAT infrastructure, migrate the data, and train staff.
- A SaaS / cloud-hosted LMS was considered but not carried forward in this engagement: it is inconsistent with YAT's current on-premises operating model and is noted as a future option.

5.3 Evaluation method

Each option is evaluated with a five-year cost-benefit analysis (§6), an intangibles comparison covering support, compliance, reliability and maintenance burden (§7.1), and a risk register (§7.2). Because the decision turns largely on supportability and compliance rather than on cost alone, the intangibles carry significant weight.

5.4 Initial impact and difficulty assessment

	Option A — Remediate GrayBoard	Option B — Replace (DOODLE)
Strategic impact	Defers the problem; remains a dead-end platform	Resolves it; supported, compliant, future-ready
Implementation difficulty	Low — incremental work on a known system	Moderate — selection, deployment, data migration, training
Headline pros	Lower upfront cost; familiar platform	Vendor support + roadmap; compliant; lower ongoing maintenance
Headline cons	Unsupported, non-compliant, replacement still looms	Higher Year-1 cost; migration and change effort

6. Cost-Benefit Analysis

A five-year cost comparison of the two options (AUD ex GST, 2022 price levels; inflation excluded for comparison clarity). Detailed line items are in Appendix 1.

6.1 Assumptions

Assumption	Value	Source
GrayBoard in-house maintenance effort	~0.4 FTE / year	YAT ICT staffing records, FY2021–22
Fully-loaded ICT staff cost	\$110,000 / year	YAT Finance, 2022 salary bands
DOODLE licensing and vendor support	\$18,400 / year	DOODLE vendor quotation, 4 March 2022
GrayBoard server refresh under Option A	\$55,000 one-off	YAT ICT hardware quotation, February 2022
Cost of LMS downtime during teaching hours	\$620 / hour	YAT Finance estimate — teaching hours lost × delivery cost, 2022
Price basis	2022 levels, AUD ex GST, inflation excluded	AccentLoitte costing convention for five-year comparisons

Each figure is attributed so the board can see what the comparison rests on and challenge any assumption it disagrees with.

6.2 Option A — 5-year summary

	Year 1	Year 2	Year 3	Year 4	Year 5	5-year
One-off remediation	\$55,000	—	—	—	—	\$55,000
Recurring (maintenance)	\$67,000	\$67,000	\$67,000	\$67,000	\$67,000	\$335,000
Annual total	\$122,000	\$67,000	\$67,000	\$67,000	\$67,000	\$390,000

6.3 Option B — 5-year summary

	Year 1	Year 2	Year 3	Year 4	Year 5	5-year
One-off project	\$160,000	—	—	—	—	\$160,000
Recurring (run + support)	(in project)	\$73,400	\$73,400	\$73,400	\$73,400	\$293,600
Annual total	\$160,000	\$73,400	\$73,400	\$73,400	\$73,400	\$453,600

6.4 Quantified benefit — avoided downtime

GrayBoard's unplanned outages over FY2021–22 totalled approximately 71 hours during teaching hours. At the \$620/hour cost of lost teaching delivery in §6.1, that is about \$44,000 a year in disruption YAT is absorbing today. A supported platform meeting the $\geq 99\%$ availability target would hold unplanned downtime under about 18 teaching hours a year, avoiding roughly \$33,000 of that annually — some \$165,000 over the five-year horizon.

This benefit is stated separately from the cost tables rather than netted off them: it is an avoided cost YAT currently absorbs across the teaching timetable, not a line item in either option's budget. Counted against the \$63,600 five-year cost difference, it more than covers the gap on its own.

6.5 Comparison summary

	Option A	Option B	Delta (B – A)
One-off (Year 1)	\$55,000	\$160,000	+\$105,000
Recurring (per year)	\$67,000	\$73,400	+\$6,400
5-year direct cost	\$390,000	\$453,600	+\$63,600

Option B costs more on every measure — ~\$105,000 more upfront and ~\$6,400 more per year, ~\$63,600 more over five years. The case for it is therefore not cost but supportability, compliance and performance: Option A's lower spend buys only a temporary reprieve on an unsupported, non-compliant platform that must still be replaced. Option A's cost advantage also rests on the maintenance effort assumed in §6.1 holding steady, which on this platform's trajectory it will not.

7. Risk and Impact Assessment

7.1 Intangibles comparison

Factor	Option A — Remediate GrayBoard	Option B — Replace (DOODLE)
Vendor support & roadmap	None — bespoke dead-end	Active vendor support + roadmap
Accessibility (WCAG 2.1 AA / DDA)	Fragile retrofit	Native conformance
Availability ($\geq 99\%$)	Degrading	Meets target on supported stack
Peak-load performance	Degrades at term-end	Match/exceed; provisioned to peak
YAT ICT maintenance burden	High (in-house bespoke)	Reduced (supported product)

Future-proofing	Replacement still looms	Current platform; cloud-ready later
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7.2 Risk register (recommended option)

Risk	Likelihood	Impact	Mitigation
Data loss/corruption in migration	Low	High	Verified ETL with record reconciliation; parallel run; tested backups
Cutover disrupts teaching	Medium	High	Cutover in the Dec–Jan break; rollback plan; staged go-live
Low user adoption / support spike	Medium	Medium	Role-based training before cutover; stabilisation support period
Integration gaps (AD / 0365 / AVETMISS)	Medium	Medium	Integration testing during parallel run before cutover
Schedule slips into a teaching term	Low	Medium	Phase gates; the fixed Dec–Jan cutover window

8. Recommendation

Recommended option: Option B — replace GrayBoard with DOODLE, deployed on-premises. Although ~\$63,600 more over five years, it is the only option that delivers a supported, accessible, performant LMS with a product roadmap, reduces YAT's ongoing maintenance burden, and preserves the student-record history — while Option A merely defers an unavoidable replacement on an unsupported, non-compliant platform. This recommendation drives the phased action plan below.

9. Action Plan

9.1 Prioritised changes

#	Phase	Priority rationale
1	Selection & implementation plan	Confirm the product and the deployment/migration approach first
2	Deployment & data migration	Stand up DOODLE, migrate and verify records, run in parallel
3	Stabilisation & closure	Support, training, as-built docs, knowledge transfer, closure

9.2 Implementation schedule

Phase	Activities	Window	Dependencies	Owner
1	Requirements, market evaluation, implementation plan	2022 H1	—	AccentLoitte
2	Infra provisioning, DOODLE install, ETL migration, parallel run, cutover	Dec 2022 – Jan 2023 break	Phase 1 approval	AccentLoitte + YAT ICT
3	Stabilisation support, training, as-built docs, closure pack	Jan 2023	Phase 2 cutover	AccentLoitte + YAT ICT

Staff training is delivered before cutover to contain the post-cutover support burden, and the cutover is scheduled into the December–January teaching break.

9.3 Standards, targets, and success metrics

Aspect	Standard / target	How measured
Availability	≥ 99%	LMS uptime monitoring over the stabilisation period
Accessibility	WCAG 2.1 AA	Accessibility audit of the deployed LMS
Record preservation	100% reconciliation	Post-migration record counts vs pre-migration; sample fidelity checks
Vendor support	Sev-1 ≤ 4h (teaching hours)	Support contract terms; incident logs

9.4 Implementation methods

Deployment follows a staged method rather than a single switch. DOODLE is stood up on new infrastructure alongside the running GrayBoard platform; records are migrated by a verified ETL process with record-count reconciliation and sample fidelity checks; the two platforms then run in parallel while integrations to Active Directory, Office 365 and AVETMISS reporting are tested against live data. Cutover happens only once parallel running is clean, and GrayBoard is retained in a read-only state through the stabilisation period so there is a rollback path.

9.5 Alignment with change-management procedure

The cutover is a high-risk change requiring Change Advisory Board endorsement and YAT senior-management sign-off; the Phase 3 Closure Pack is signed off under the YAT Change Management Procedure.

10. Next Steps and Decision Requested

We ask the board to: (1) approve Option B — replacing GrayBoard with DOODLE on-premises; (2) approve the phased implementation plan in §9; and (3) authorise the Year-1 budget envelope (≈\$160,000). The high-risk cutover change request is submitted at the Phase 2 gate per the change management procedure.

11. Feedback

Feedback received on this Business Case during review, and how each item was addressed before approval.

From	Date	Feedback	How it was addressed
Sam Walker, YAT ICT Manager	3 May 2022	The five-year comparison shows Option B costing more but does not put a number on what the current outages are costing YAT.	Added \$6.4 quantifying avoided downtime at ~\$33,000 a year, and the \$620/hour basis for it in §6.1.
Robin Ng, Chief Financial Officer	9 May 2022	Where do the staffing and licensing figures come from?	Added a Source column to the §6.1 assumptions table attributing every

			figure.
Sam Walker, YAT ICT Manager	16 May 2022	The plan needs to be explicit that there is a way back if cutover goes badly.	Expanded §9.4 to state the parallel-run method and the read-only GrayBoard rollback path through stabilisation.

Sign-off

Role	Name	Date	Signature
Prepared by	AccentLoitte Consultant	12 May 2022	(signed)
Reviewed by	Jamie O'Donnell (AccentLoitte Senior Consultant)	13 May 2022	(signed)
Approved by	Sam Walker (YAT ICT Manager)	20 May 2022	(signed)

Appendix 1 — Cost Detail

A1.1 Option A — Remediate and retain GrayBoard

One-off remediation (Year 1):

Item	Cost
Accessibility retrofit to WCAG 2.1 AA	\$35,000
Legacy server hardware refresh	\$20,000
Total one-off	\$55,000

Recurring (per year):

Item	Annual cost
In-house bespoke maintenance (~0.4 FTE)	\$44,000
Incident response (~0.10 FTE, rising)	\$11,000
Hosting, facilities, backup	\$12,000
Total recurring per year	\$67,000

A1.2 Option B — Replace with DOODLE (on-premises)

One-off project (Year 1):

Item	Cost
AccentLoitte fees (Phases 1-3: selection, deployment, migration, training, closure)	\$95,000
On-premises infrastructure (server, storage, UPS, backup)	\$35,000
YAT ICT staff time (oversight, integration, acceptance, parallel run)	\$22,000
Training delivery	\$8,000
Total Year-1 project	\$160,000

Recurring (per year, Years 2-5):

Item	Annual cost
DOODLE vendor support & maintenance contract	\$30,000
Windows Server licensing + antivirus + monitoring	\$3,800
Power, facilities, server-room allocation	\$6,700
Backup (media, offsite storage, software maintenance)	\$5,400
YAT ICT staff (admin 0.20 FTE + incident 0.05 FTE)	\$27,500
Total recurring per year	\$73,400

Appendix 2 — LMS Market Evaluation (summary)

Candidate replacement platforms were scored against the §5.1 requirements. DOODLE was selected on the balance of functional fit, total cost of ownership, RTO-sector fit, accessibility conformance, and vendor longevity.

Criterion	DOODLE (selected)	Alternative 1	Alternative 2
Functional fit to YAT requirements	Strong	Strong	Moderate
On-prem deployment supported	Yes	Yes	SaaS only
WCAG 2.1 AA conformance	Yes	Partial	Yes
AD / O365 / AVETMISS integration	Yes	Yes	Limited
5-year total cost of ownership	Competitive	Higher	Lower (but SaaS — out of scope)
Vendor longevity & roadmap	Strong	Moderate	Strong